

The University of Azad Jammu & Kashmir, Muzaffarabad

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Software Design & Architecture (SDA)

Activity: Hospital Appointment System

1. List 6–10 Candidate Conceptual Classes

A Hospital Appointment System manages patients, doctors, appointments, medical records, and billing.

The following are the candidate conceptual classes:

1. **Patient** – Stores information about patients.
2. **Doctor** – Stores information about doctors and their specializations.
3. **Appointment** – Manages appointments between patients and doctors.
4. **MedicalRecord** – Stores patients' medical history and diagnoses.
5. **Bill** – Maintains billing information for hospital services.
6. **Payment** – Records payments made by patients.
7. **Prescription** – Stores medicines prescribed by doctors.
8. **Department** – Represents hospital departments, such as Cardiology and Neurology.
9. **Receptionist** – Manages patient registration and appointments.
10. **Schedule** – Maintains doctors' available dates and times.

2. Attributes and Methods of Three Important Classes

A. Patient Class

Attributes (Data/Properties):

- patientID
- name
- age
- gender
- phoneNumber
- address

Methods (Behaviors):

- registerPatient()
- updatePatientDetails()
- bookAppointment()
- viewMedicalRecord()

B. Doctor Class

Attributes (Data/Properties):

- doctorID
- name
- specialization
- phoneNumber
- availableDays

Methods (Behaviors):

- viewAppointments()
- updateSchedule()
- examinePatient()
- prescribeMedicine()

C. Appointment Class

Attributes (Data/Properties):

- appointmentID
- patientID
- doctorID
- appointmentDate
- appointmentTime
- status

Methods (Behaviors):

- bookAppointment()
- cancelAppointment()
- rescheduleAppointment()
- confirmAppointment()

3. Abstraction Decisions

In the Hospital Appointment System, I will focus on the information necessary to manage patients, doctors, appointments, medical records, and billing.

I will ignore unnecessary real-world details, such as a patient's height, weight, clothing, and personal hobbies, because they are not essential for booking appointments.

I will also exclude details about the hospital building, such as room colors and furniture designs, because they do not affect appointment management.

This abstraction makes the system simpler, easier to understand, and more efficient.

4. One Invented Software Object

AppointmentController

The AppointmentController is an invented software object that helps manage the appointment process.

It does not represent a physical object in the real world. Instead, it connects patients, doctors, and appointments within the software system.

Attributes:

- controllerID
- appointmentStatus

Methods:

- validateAppointment()
- checkDoctorAvailability()
- confirmBooking()
- sendAppointmentNotification()

Purpose: The AppointmentController ensures that appointments are valid, doctors are available, and booking requests are processed correctly.

Conclusion

The Hospital Appointment System uses conceptual classes to represent important entities and their responsibilities. Attributes describe the data of each class, while methods define its behavior. Abstraction removes unnecessary details, and the invented software object AppointmentController helps coordinate the appointment process.